CLIMATE ECONOMICS, GREEN FINANCE AND CIRCULAR ECONOMY

EXTERNALITY-TO-TRANSITION RAIL → MITIGATION-ADAPTATION SPLIT → CARBON-VALUE TRIANGLE → PAT-TO-CCTS BOUNDARY → CLIMATE-RISK CHANNELS → GREEN-FINANCE CREDIBILITY CHAIN

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

◆ PRIMARY CORE FLOW ◇ SUBORDINATE EXTRA ◆ APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • active generation g1 • explicit approval of this exact generation is required • all prior artifacts unchanged

00

STAGE 00

EXTERNALITY-TO-TRANSITION RAIL

EXTERNALITY-TO-TRANSITION RAIL · RESOURCE USE + EMISSIONS · UNPRICED SOCIAL COST · PRICE + RULE + FINANCE · LOW-CARBON RESILIENT OUTCOME

RESOURCE USE + EMISSIONS → UNPRICED SOCIAL COST → PRICE + RULE + FINANCE → LOW-CARBON RESILIENT OUTCOME

CONCEPT / ACCOUNTING BASIS
- RESOURCE USE + EMISSIONS
- UNPRICED SOCIAL COST

TRANSMISSION / RECONCILIATION
- PRICE + RULE + FINANCE
- LOW-CARBON RESILIENT OUTCOME

DISTRIBUTIONAL / STABILITY LIMIT
- RESOURCE USE + EMISSIONS
- UNPRICED SOCIAL COST

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PRICE + RULE + FINANCE; LOW-CARBON RESILIENT OUTCOME.

01

STAGE 01

MITIGATION-ADAPTATION SPLIT

MITIGATION-ADAPTATION SPLIT · EMISSIONS + SINKS · EXPOSURE + VULNERABILITY · FINANCE MODELS DIFFER · BOTH REQUIRED; NEITHER SUBSTITUTES

MEASURE / INSTITUTION	DECISIVE DISTINCTION	VINTAGE / LEGAL LIMIT
MITIGATION → emissions + sinks	FINANCE MODELS differ	MITIGATION → emissions + sinks
ADAPTATION → exposure + vulnerability	BOTH required; neither substitutes	ADAPTATION → exposure + vulnerability

MEASURE / INSTITUTION
- MITIGATION → emissions + sinks
- ADAPTATION → exposure + vulnerability

DECISIVE DISTINCTION
- FINANCE MODELS differ
- BOTH required; neither substitutes

VINTAGE / LEGAL LIMIT
- MITIGATION → emissions + sinks
- ADAPTATION → exposure + vulnerability

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ADAPTATION → exposure + vulnerability; FINANCE MODELS differ; BOTH required; neither substitutes.

02

STAGE 02

CARBON-VALUE TRIANGLE

CARBON-VALUE TRIANGLE · DAMAGE VALUATION · CARBON TAX · STATUTORY PRICE · TRADING PRICE · MARKET OUTCOME · THREE VALUES MAY DIVERGE

SCC → damage valuation

CARBON TAX → statutory price → TRADING PRICE → market outcome → THREE VALUES may diverge

CONCEPT / ACCOUNTING BASIS
- SCC → damage valuation
- CARBON TAX → statutory price

TRANSMISSION / RECONCILIATION
- TRADING PRICE → market outcome
- THREE VALUES may diverge

DISTRIBUTIONAL / STABILITY LIMIT
- SCC → damage valuation
- CARBON TAX → statutory price

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRADING PRICE → market outcome; THREE VALUES may diverge.

03

STAGE 03

PAT-TO-CCTS BOUNDARY

PAT-TO-CCTS BOUNDARY · ENERGY INTENSITY · GHG COMPLIANCE ARCHITECTURE · SECTOR + YEAR · DATED NOTIFICATION · PHASED STATUS · ECONOMY-WIDE MARKET

PAT → energy intensity

CCTS → GHG compliance architecture → SECTOR + YEAR → dated notification → PHASED STATUS != economy-wide market

MEASURE / INSTITUTION
- PAT → energy intensity
- CCTS → GHG compliance architecture

DECISIVE DISTINCTION
- SECTOR + YEAR → dated notification
- PHASED STATUS != economy-wide market

VINTAGE / LEGAL LIMIT
- PAT → energy intensity
- CCTS → GHG compliance architecture

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SECTOR + YEAR → dated notification; PHASED STATUS != economy-wide market.

04

STAGE 04

CLIMATE-RISK CHANNELS

CLIMATE-RISK CHANNELS · HAZARD EXPOSURE · POLICY + TECHNOLOGY · CREDIT + MARKET + OPERATIONAL · HOUSEHOLD + FISCAL TRANSMISSION

PHYSICAL → hazard exposure

TRANSITION → policy + technology → CREDIT + MARKET + OPERATIONAL → HOUSEHOLD + FISCAL transmission

CONCEPT / ACCOUNTING BASIS
- PHYSICAL → hazard exposure
- TRANSITION → policy + technology

TRANSMISSION / RECONCILIATION
- CREDIT + MARKET + OPERATIONAL
- HOUSEHOLD + FISCAL transmission

DISTRIBUTIONAL / STABILITY LIMIT
- PHYSICAL → hazard exposure
- TRANSITION → policy + technology

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CREDIT + MARKET + OPERATIONAL; HOUSEHOLD + FISCAL transmission.

05

STAGE 05

GREEN-FINANCE CREDIBILITY CHAIN

GREEN-FINANCE CREDIBILITY CHAIN · OUTPUT + OUTCOME METRIC · ASSURANCE + ADDITIONALITY · ELIGIBILITY / TAXONOMY

ELIGIBILITY / TAXONOMY → OUTPUT + OUTCOME METRIC → ASSURANCE + ADDITIONALITY

CONCEPT / ACCOUNTING BASIS
- ELIGIBILITY / TAXONOMY
- OUTPUT + OUTCOME METRIC

TRANSMISSION / RECONCILIATION
- ASSURANCE + ADDITIONALITY
- ELIGIBILITY / TAXONOMY

DISTRIBUTIONAL / STABILITY LIMIT
- OUTPUT + OUTCOME METRIC
- ASSURANCE + ADDITIONALITY

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ALLOCATION; OUTPUT + OUTCOME METRIC; ASSURANCE + ADDITIONALITY.

06

STAGE 06

BOND-USE BOUNDARY

BOND-USE BOUNDARY · GREEN BOND · ENVIRONMENTAL USES · SUSTAINABILITY BOND · GREEN + SOCIAL · SOVEREIGN ISSUE · EARMARKED BORROWING · COMPLETED VERIFIED OUTCOME

MEASURE / INSTITUTION	DECISIVE DISTINCTION	VINTAGE / LEGAL LIMIT
GREEN BOND → environmental uses	SOVEREIGN ISSUE → earmarked borrowing	GREEN BOND → environmental uses
SUSTAINABILITY BOND → green + social	LABEL != completed verified outcome	SUSTAINABILITY BOND → green + social

MEASURE / INSTITUTION
- GREEN BOND → environmental uses
- SUSTAINABILITY BOND → green + social

DECISIVE DISTINCTION
- SOVEREIGN ISSUE → earmarked borrowing
- LABEL != completed verified outcome

VINTAGE / LEGAL LIMIT
- GREEN BOND → environmental uses
- SUSTAINABILITY BOND → green + social

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SOVEREIGN ISSUE → earmarked borrowing; LABEL != completed verified outcome.

07

STAGE 07

CIRCULAR-VALUE LADDER

CIRCULAR-VALUE LADDER · REDUCE + REDESIGN · REUSE + REPAIR · REFURBISH + REMANUFACTURE · RECYCLE RESIDUAL MATERIALS

REDUCE + REDESIGN

REUSE + REPAIR → REFURBISH + REMANUFACTURE → RECYCLE residual materials

CONCEPT / ACCOUNTING BASIS
- REDUCE + REDESIGN
- REUSE + REPAIR

TRANSMISSION / RECONCILIATION
- REFURBISH + REMANUFACTURE
- RECYCLE residual materials

DISTRIBUTIONAL / STABILITY LIMIT
- REDUCE + REDESIGN
- REUSE + REPAIR

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: REUSE + REPAIR; REFURBISH + REMANUFACTURE; RECYCLE residual materials.

08

STAGE 08

EPR RESULTS CHAIN

EPR RESULTS CHAIN · PRODUCER OBLIGATION · CERTIFICATE + VERIFICATION · ACTUAL MATERIAL OUTCOME

CONCEPT / ACCOUNTING BASIS
- PRODUCER OBLIGATION
- COLLECTION / PROCESSING

TRANSMISSION / RECONCILIATION
- CERTIFICATE + VERIFICATION
- ACTUAL MATERIAL OUTCOME

DISTRIBUTIONAL / STABILITY LIMIT
- PRODUCER OBLIGATION
- COLLECTION / PROCESSING

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: COLLECTION / PROCESSING; CERTIFICATE + VERIFICATION; ACTUAL MATERIAL OUTCOME.

09

STAGE 09

JUST-TRANSITION MAP

JUST-TRANSITION MAP · WORKERS + SKILLS · REGIONS + REVENUE · CONSUMERS + AFFORDABILITY · ENERGY ACCESS + SOCIAL PROTECTION

WORKERS + SKILLS → REGIONS + REVENUE → CONSUMERS + AFFORDABILITY → ENERGY ACCESS + SOCIAL PROTECTION

CONCEPT / ACCOUNTING BASIS
- WORKERS + SKILLS
- REGIONS + REVENUE

TRANSMISSION / RECONCILIATION
- CONSUMERS + AFFORDABILITY
- ENERGY ACCESS + SOCIAL PROTECTION

DISTRIBUTIONAL / STABILITY LIMIT
- WORKERS + SKILLS
- REGIONS + REVENUE

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CONSUMERS + AFFORDABILITY; ENERGY ACCESS + SOCIAL PROTECTION.

10

STAGE 10

TRADE-CLIMATE BOUNDARY

TRADE-CLIMATE BOUNDARY · EU CBAM · EXTERNAL-IMPORT RULE · INDIA CCTS · DOMESTIC SYSTEM · UNSETTLED UNLESS VERIFIED · TRADE PRESSURE · POLICY IDENTITY

EU CBAM → external import rule

INDIA CCTS / MRV → domestic system → RECOGNITION → unsettled unless verified → TRADE PRESSURE != policy identity

MEASURE / INSTITUTION
- EU CBAM → external import rule
- INDIA CCTS / MRV → domestic system

DECISIVE DISTINCTION
- RECOGNITION → unsettled unless verified
- TRADE PRESSURE != policy identity

VINTAGE / LEGAL LIMIT
- EU CBAM → external import rule
- INDIA CCTS / MRV → domestic system

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RECOGNITION → unsettled unless verified; TRADE PRESSURE != policy identity.

11

STAGE 11

CLIMATE-ECONOMICS ANSWER SPINE

CLIMATE-ECONOMICS ANSWER SPINE · DEFINE METRIC + STATUS · TRACE PRICE + FINANCE + INNOVATION · TEST DISTRIBUTION + ADDITIONALITY · CONCLUDE CREDIBLE JUST TRANSITION

DEFINE metric + status → TRACE price + finance + innovation → TEST distribution + additionality → CONCLUDE credible just transition

CONCEPT / ACCOUNTING BASIS
- DEFINE metric + status
- TRACE price + finance + innovation

TRANSMISSION / RECONCILIATION
- TEST distribution + additionality
- CONCLUDE credible just transition

DISTRIBUTIONAL / STABILITY LIMIT
- DEFINE metric + status
- TRACE price + finance + innovation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TEST distribution + additionality; CONCLUDE credible just transition.

E

EXTRA

SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · THE SOCIAL COST OF CARBON VALUES MARGINAL CLIMATE DAMAGE · GREEN TAXONOMIES REDUCE INFORMATION ASYMMETRY WHILE RIGID LABELS MAY · CLIMATE STRESS TESTING EXPLORES SCENARIO EXPOSURE RATHER THAN PREDICTING · BORDER CARBON MEASURES CONNECT DOMESTIC CLIMATE AMBITION WITH TRADE · CIRCULAR BUSINESS MODELS INCLUDE PRODUCT-AS-SERVICE, INDUSTRIAL SYMBIOSIS, REPAIR · ADAPTATION FINANCE HAS PUBLIC-GOOD CHARACTERISTICS AND WEAKER DIRECT REVENUE · MINISTRY OF ENVIRONMENT, FOREST AND CLIMATE CHANGE

OPTIONAL MODEL / MEASUREMENT DEPTH
- CAUTION: The social cost of carbon values marginal climate damage but depends on discounting and
- CAUTION: Green taxonomies reduce information asymmetry while rigid labels may exclude credible
- CAUTION: Climate stress testing explores scenario exposure rather than predicting one exact
- CAUTION: Border carbon measures connect domestic climate ambition with trade competitiveness and

VINTAGE, BASKET OR LEGAL LIMIT
- CAUTION: Circular business models include product-as-service, industrial symbiosis, repair
- CAUTION: Adaptation finance has public-good characteristics and weaker direct revenue streams
- Ministry of Environment, Forest and Climate Change: coordinates climate policy and
- Ministry of New and Renewable Energy and power-sector institutions: support clean

COMPARATIVE POLICY NUANCE
- RBI and SEBI: shape climate-risk disclosure, green debt and financial-sector
- Urban local bodies and pollution-control systems: implement waste, recycling and
- CAUTION: A carbon price can reduce emissions efficiently but raise household energy costs unless
- CAUTION: A green bond finances defined environmental uses; a sustainability bond can combine

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels